

Adaptive Spatial-Temporal Graph Convolutional Network (AAGCN)

AAGCN Macro Backbone

Input Kinematic Stream

$[B, C = 3, T = 32, V = 13]$

Data BatchNorm2d(3)

Joint-level normalisation

AAGCN Stage 1

Channels: $3 \rightarrow 48$, Stride: 1

AAGCN Stage 2

Channels: $48 \rightarrow 96$,
Stride: 2 ($T : 32 \rightarrow 16$)

AAGCN Stage 3

Channels: $96 \rightarrow 150$,
Stride: 2 ($T : 16 \rightarrow 8$)

GAP over (V, T)

Output: $[B, 150]$

Linear Classifier Head

Linear($150 \rightarrow 22$) $\rightarrow$ Softmax
Yields $P_m \in \Delta^{21}$

Adaptive Graph Conv (Spatial-Temporal Block)

Branch 1

$A_{\text{topo}} \odot M$
(Anatomy)

Branch 2

Matrix B
(Global)

Branch 3

$C(X)$
Attention
(Dynamic)

Unified Adaptive Adjacency Matrix

$$A_{\text{total}} = (A_{\text{topo}} \odot M) + B + C(X)$$

Spatial Graph Message Passing

$$X_g = A_{\text{total}} \cdot X \rightarrow \text{Conv2d}(1 \times 1) \rightarrow \text{BN} \rightarrow \text{GELU}$$

Multi-Scale Temporal Conv (MS-TCN)

$$\text{Conv2d}(9 \times 1, \text{stride} = s) \rightarrow \text{BN} \rightarrow \text{Dropout}(0.2)$$

Residual Shortcut Connection (+)

$$\text{Shortcut: } \text{Conv2d}(1 \times 1, \text{stride} = s) + \text{BN} \rightarrow (+) \text{ Add} \rightarrow \text{GELU}$$

Block Output Feature Map

Passed to next stage or GAP